Live Non-Isolated Sign Language Recognition

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Introduction

Problem Statement:

- Computers struggle to quickly translate sign language characters and gestures into understandable and complete sentences. Thus, we aim to solve the problem of live non-isolated sign language interpretation, both in terms of real-time translation for spelling out words (simple) and recognizing gestures (hard) from the ASL vocabulary while balancing speed and accuracy to suit different real-world applications. For both tasks, the input to the model is a live video stream that is either downscaled to 224x224 or processed with Google's MediaPipe algorithm. The output of the model is initially a character or word from a pre-specified vocabulary, which is later translated into a coherent sentence.

Overview:

- Video/gesture recognition using frame preprocessing with Google's MediaPipe offers the best performance for live video-to-text translations given its lightweight nature, ideal for quick identification of common words.
- Character recognition is best solved by a Vision Transformer (ViT) model enhanced with transfer learning if time is no issue; for efficiency and high frame-rate videos, the Isolated MediaPipe Linear Model excels.

Datasets

Character Recognition:

- 87,000 images of ASL finger spelling across 26 letters, augmented with random stretches, flips, rotations, translations, and significant noise.

Gesture Recognition:

- 12,000 videos for 2,000 words from the WLASL dataset
- 94,744 videos for 250 words from the MediaPipe GISLR competition dataset.

Preprocessing:

- Vit/ViVit models: videos were divided into 16 frames, converted to grayscale, and resized to 224x224 dimensions; images were also resized to 224x224.
- MediaPipe models: Google's MediaPipe algorithm was used to generate a set of "landmarks" for hands, lips, and pose; videos were resized to 64 frames

ViT Letter Recognition

- **Sub-Image partitioning:** Images are broken down into P smaller C x C dimension panels forming a grid, panels are passed through a CNN before being fed into the transformers encoder.

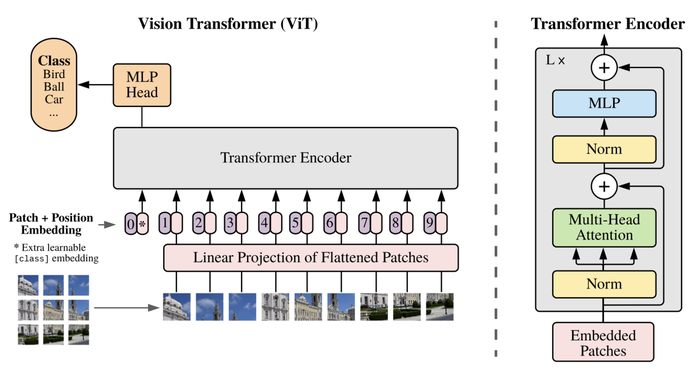

Figure 1. Architecture of the Vision Transformer (ViT) model.

Isolated MediaPipe Character Recognition

- **Dimensional Reduction:** From our MediaPipe tensors, either three fully connected layers or three 1D convolutional layers are used for feature extraction
- **Optimization Techniques:** Constant normalization and dropout neurons enable high test performance.

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Figure 2. Sample Class Visualization of MediaPipe Isolated Character Best Model

ViViT Video Classification

- **Spatiotemporal Attention and Patch Embedding:** captures interactions between panels in the same time frame.
- **Temporal Self Attention Mechanism:** Analyzes patterns between frames of different time stamps.
- **One depth channel** to cut down on storage space needed for video tensors

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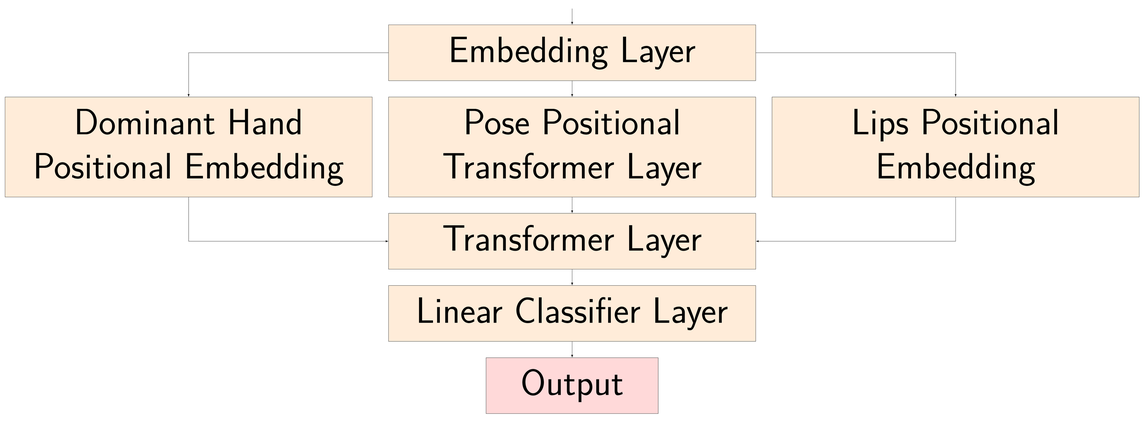
Figure 3. Architecture of the Video Vision Transfomer (ViViT) model.

MediaPipe Gesture Recognition (FFIMpGIM)



- **Positional Embedding:** unique vectors for each temporal position in the input sequence, enhancing the model's understanding of frame order.
- **Feature-Specific Embedding:** position, velocity, and acceleration vectors for lips, hands, and pose with adjustable learning weights

Results

Table 1. Table 1: Models vs Baseline for Character Identification

Model	Test Accuracy
Bantupalli (baseline)	91%
Raval (baseline)	83%
Paul (baseline)	93.8%
Deepali (baseline)	95.31%
MP Linear NN	98.76%
MP CNN	98.57%
MP + Norm	88.9%
ViT	99.84%

Table 2. Table 2: Models vs Baseline for Gesture Identification

Model [# of classes]	Test Accuracy
Paul [3] (baseline)	94.3%
Luna [13] (baseline)	62.79%
Hoyso [250] (baseline)	80%
ViViT [400]	28.9%
LSTM: flattened [400]	1.4%
Transf: flattened [400]	1.2%
MP: CustEnc [250]	61.83%
MP: CustEnc + va [250]	73.56%

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Figure 4. Training Loss, Validation Loss, and Accuracy of the Linear MP Model and FFIMpGIM

Conclusions and Future Work

Main Findings

- Models with raw video inputs instead of hand coordinates perform too slow for real time video applications
- Classifiers are more robust with data augmentation that randomly scales, rotates, and manipulates training data
- Computer vision models can be readily appended with word models for fluent translations

Future Work

- Translation word models tailored for computer vision applications
- Synthesizing multiple models to maximize accurate translations